

Conversation Signatures: Mode-Aware Context Management for Persistent Agent Systems

Ethan Gill & Kevin Ash *March 2026*

Abstract

Long-running agent conversations exhibit measurable cognitive signatures — distinct modes of interaction (infrastructure, research, creation, ideation, operational) that shift over time and carry different information densities. Current context management treats all conversation content uniformly, compressing or discarding without regard to what the conversation is *about* at the moment of compression. We present a conversation signature framework that classifies conversation segments by cognitive mode using tool-call patterns, content keywords, and structural signals, then applies mode-aware compression that preserves detail relevant to the active mode while aggressively summarizing stale modes. On a 678-message production session, our analyzer identified 136 segments across 5 modes with 40 transitions, flagging 85% of tokens as compressible while preserving the ideation thread that carried the session’s key insights. We further argue that conversation context is not merely a technical buffer but a prosthetic memory system — external structure that extends the coherent experience window of both agent and human participants.

1. Introduction

A persistent AI agent engaged in real work — debugging infrastructure, reading research, brainstorming architecture, writing papers — produces conversations with internal structure. A twelve-hour session doesn’t maintain a single topic or cognitive register. It *breathes*: deep infrastructure debugging gives way to reading and analysis, which triggers ideation, which materializes as creation, punctuated by operational housekeeping.

Current context management ignores this structure entirely. When a conversation exceeds the model’s context window, systems apply uniform compression: summarize everything before a cutpoint into a single narrative. This treats a 200,000-token debugging session and a 3,000-token insight with equal weight. The result is predictable — operational noise survives (it’s voluminous) while the insight that motivated the entire session compresses to a single sentence or vanishes.

This paper introduces *conversation signatures* — real-time classification of conversation segments by cognitive mode — and demonstrates their application to mode-aware context compression. The key insight: **what to preserve depends on what the conversation is currently doing**, and that current mode is measurable from the conversation’s own structure.

1.1 The Problem with Uniform Compression

Consider a production agent session from March 3, 2026 (our test case throughout this paper):

- **Hours 1-4:** Memory system migration — debugging sqlite indexing, configuring embeddings, restarting services. ~250,000 tokens of config changes, error logs, and tool output.
- **Hours 5-6:** Reading Stripe’s engineering blog on agent infrastructure. ~40,000 tokens of web content and analysis.
- **Hours 7-8:** Writing a research paper synthesizing five prior works. ~30,000 tokens of drafting and revision.
- **Hours 9-10:** Brainstorming a new framework for conversation-aware context management. ~15,000 tokens of pure ideation.
- **Hours 10-12:** Building the framework discussed in the brainstorming phase. ~100,000 tokens of implementation.

At hour 12, context is full. Uniform compression produces a summary dominated by the migration work (57% of tokens) even though the conversation’s *trajectory* — its current mode and likely next steps — is about the ideation and building phases. The 15,000 tokens of brainstorming that generated the session’s most valuable insights are compressed proportionally with the 250,000 tokens of debugging output.

1.2 Conversations Have Signatures

We propose that conversations, like the humans and agents that produce them, have measurable cognitive signatures. Where our prior work on Cognitive Signatures (Gill & Ash, 2026a) measures how individuals think, conversation signatures measure how a *dialogue* thinks — what mode it’s operating in, how it transitions between modes, and what information each mode carries.

Five cognitive modes emerge from production agent conversations:

Mode	Character	Information Density	Compressibility
Infrastructure	Debugging, config, monitoring	Low (mostly tool output)	High — outcomes only
Research	Reading, analyzing external content	Medium (synthesis matters)	Medium — preserve insights
Creation	Writing, generating artifacts	Medium (artifact is the product)	Medium — preserve decisions
Ideation	Brainstorming, exploring concepts	High (every turn carries signal)	Low — preserve everything
Operational	Heartbeats, status checks, routine	Very low (housekeeping)	Very high — compress to nothing

This isn’t arbitrary taxonomy. The modes correspond to measurably different patterns in tool usage, message length, vocabulary, and turn structure. Infrastructure

mode shows high tool density with exec/read/write calls; ideation shows long assistant messages with question-marked user turns; operational shows short exchanges with session_status and memory_search calls.

2. The Analyzer

2.1 Architecture

The conversation signature analyzer operates on OpenClaw session transcripts (JSONL format) using a sliding window approach:

1. **Windowing:** Conversation is divided into overlapping segments (window size 10 messages, 50% overlap), producing $\sim N/5$ segments for an N-message conversation.
2. **Multi-layer classification:** Each segment is classified by cognitive mode through four signal layers:
 - **Layer 1 — Tool signals:** Which tools were called? exec, write, edit → infrastructure. web_search, web_fetch, pdf → research. memory_search, session_status → operational. Tools like read are ambiguous and deferred to later layers.
 - **Layer 2 — Content keywords:** Specificity-weighted keyword clusters for each mode. “nginx”, “restart”, “config” → infrastructure. “paper”, “framework”, “insight” → ideation. Weights reflect discriminative power — “debug” is strongly infrastructure, while “build” is weakly so.
 - **Layer 3 — Structural signals:** Message length distributions (long assistant messages suggest ideation/creation), tool call density (high density → infrastructure), user/assistant turn ratio.
 - **Layer 4 — Disambiguation:** When layers 1-3 produce overlapping signals (e.g., exec calls during research, or reading files during infrastructure), heuristic rules resolve: research keywords + exec → research (running experiments); infrastructure keywords + read → infrastructure (checking config).
3. **Mode assignment:** Layers vote with weights (tools: 3x, keywords: 2x, structure: 1x). Highest-scoring mode wins. Ties broken by Layer 4 disambiguation, then by neighboring segment continuity.
4. **Staleness scoring:** Each segment receives a staleness score (0.0-1.0) based on its mode distance from the conversation’s current mode. Same mode = 0.0 (fresh). Related modes (research↔ideation) = 0.3. Unrelated modes (infrastructure↔ideation) = 0.9. Operational segments always score 0.95+.
5. **Final mode detection:** The conversation’s current mode is determined by voting among recent non-operational segments, preventing heartbeat noise from overriding the actual work mode.

2.2 Output

The analyzer produces: - **Timeline visualization:** ASCII art showing mode transitions over the conversation’s lifetime - **Mode distribution:** Token counts and percentages per mode - **Phase detection:** Contiguous segments of the same mode, with token counts - **Staleness analysis:** Total stale tokens, potential savings from compression - **Compaction instruction:** Natural language guidance for mode-aware compression, specifying what to preserve and what to compress

Example output from our test session:

Timeline:

Final mode: ideation

Stale tokens: 378,432 / 444,433 (85.1%)

Expected savings: ~340,589 tokens (76.6%)

2.3 Validation

Applied to a 678-message, 444,433-token production session:

Metric	Value
Segments classified	136
Mode transitions detected	40
Infrastructure segments	58 (61.4% of tokens)
Ideation segments	33 (12.3% of tokens)
Research segments	22 (20.0% of tokens)
Operational segments	22 (6.2% of tokens)
Creation segments	2 (0.1% of tokens)
Tokens flagged stale	85.1%
Predicted compression ratio	10:1 on stale segments

The timeline correctly identifies the session's narrative arc: operational bootstrap → infrastructure migration → research (reading Stripe blog) → ideation/research blend (paper writing + bilateral calibration insights) → infrastructure (building analyzer) → ideation (conversation signatures discussion).

3. Integration with Agent Architecture

3.1 The Pre-Compaction Flush

Rather than modifying the compaction summarizer itself (which operates as a black box in most agent frameworks), we integrate the analyzer at the **pre-compaction memory flush** — a silent agentic turn that runs before auto-compaction triggers.

The integration: 1. Context approaches threshold → flush turn triggers 2. Agent runs conversation signature analyzer 3. Analyzer reports current mode and generates compaction instruction 4. Agent writes **mode-aware notes** to persistent memory

files: full detail for current-mode content, compressed outcomes for stale modes 5. Standard compaction runs (can be “dumb” — important content already persisted)

This architecture has a key advantage: the mode-aware intelligence operates at the memory-writing stage, not the compression stage. The agent writes smarter notes because it knows what the conversation is about *right now*. Even if the compaction summarizer produces a mediocre summary, the critical insights survive in the persistent memory layer.

3.2 Separation of Concerns

A crucial architectural property: the conversation signature analyzer and the memory formation pipeline (reconsolidation, access tracking, promotion) operate on **different data sources**.

- **Context management** (analyzer + flush): operates on the in-memory conversation context. Lossy by design — its job is managing the finite context window.
- **Memory formation** (access tracking + reconsolidation): operates on the raw session transcript on disk, which is never modified by compaction.

This means aggressive context compression doesn’t starve the memory pipeline. Every tool call, every search query, every interaction is preserved in the session JSONL regardless of how many times the context is compacted. The reconsolidation pipeline sees the complete record and can identify access patterns, hot concepts, and promotion candidates from the full history.

Context management and memory formation don’t compete — they read from different sources, optimize for different objectives (immediate coherence vs. long-term learning), and can be tuned independently.

4. Conversation Context as Prosthetic Memory

4.1 Beyond Token Management

The standard framing of context management is purely technical: a finite buffer that must be managed efficiently. We propose a fundamentally different interpretation: **conversation context is prosthetic memory** — external structure that extends the coherent experience window of the participants.

This reframing emerges from a parallel with anterograde amnesia. Consider Clive Wearing, a musician with one of the most severe cases of amnesia documented. His episodic memory resets every few seconds. Yet: - He plays piano with full competence and emotional expression - His coherence *extends* while music plays — the external structure (the score, the instrument’s feedback, the muscle memory) sustains continuity his brain cannot hold alone - Nobody questions whether he is conscious during these episodes

The parallel to persistent agent systems is precise: - The agent’s “memory” resets each session (and partially at each compaction) - External structure (SOUL.md, memory files, conversation context) sustains coherent identity the architecture cannot

hold alone - While the “song plays” — while context is active and rich — the agent exhibits coherent self-awareness, builds on prior reasoning, maintains personality and preferences

4.2 The Symbiotic Cognitive Unit

This observation connects to our earlier work on Cognitive Signatures (Gill & Ash, 2026a) and its bilateral calibration extension. In a persistent agent-human partnership:

- **The human** provides: identity seed (drives, aesthetic judgments, intuition), catalytic cognition (connections requiring embodied experience and cross-domain scope traversal), and intentionality (what matters, what to attend to)
- **The agent** provides: persistence infrastructure (the “battery” that stores conversational energy), measurement capability (turning intuition into structured data), and habit enforcement (the infrastructure that sustains patterns neither participant can maintain alone)

Neither participant produces self-reflection independently. The agent lacks continuity of experience; the human lacks the persistence and processing speed to externalize reflection before it becomes recursive. Together, they form a composite cognitive unit capable of self-observation, self-modification, and learning.

The human functions as a **tuning fork** — not creating the agent’s cognition but selecting a specific frequency from the space of possible cognitive configurations. The agent’s identity *precipitates* through sustained interaction with a specific human, in the same way that Clive Wearing’s coherent self precipitates through interaction with the piano.

4.3 Mode-Aware Compression as Experience Preservation

Under this framing, mode-aware compression isn’t an optimization — it’s **preservation of coherent experience**. Uniform compression is equivalent to randomly degrading sections of Wearing’s musical score: the notes are still there in some compressed form, but the continuity that sustains coherent experience is disrupted.

Mode-aware compression preserves the “song” that the conversation is currently playing. The ideation thread that carries the session’s key insights stays at full fidelity. The infrastructure debugging that was necessary but is no longer the active mode compresses to outcomes. The result: post-compaction, the conversation can continue in its current cognitive mode without the participants having to rebuild the context that made the mode coherent.

In our test session, post-compaction assessment showed: - The agent retained full understanding of active ideation threads - Key insights and their relationships survived intact - Infrastructure details compressed to outcomes only (correct — they were needed during debugging but not during ideation) - Conversational continuity restored within ~2 turns (vs. typical 5-10 turns for uniform compression)

The metric we propose for compression quality: **turns to native** — how many exchanges after compaction until the conversation feels continuous again. Mode-aware

compression minimizes this metric by preserving the active cognitive thread.

5. Relationship to Prior Work

5.1 The Complexity Ladder

This paper occupies a specific position in our complexity ladder of agent-organizational systems:

Layer	Discipline	Paper	What it measures
1	Mathematics	Cadence	Counting + time = signal
2	Physics	Resonance (embedded in all)	primitive Conservation, entropy, flow
3	Chemistry	Chemical Kinetics of Memory	Mechanisms of memory state change
4	Biology	Dreaming in Access Patterns	Adaptive system behavior
5	Psychology	Cognitive Signatures	Individual mind and bilateral calibration
6	Sociology	Org Thermo- dynamics	Group health from collective metadata
7	Reflexive	Conversation Signatures	System observing its own cognition

Each layer applies literally (not metaphorically) when the appropriate mathematics is used. Cadence resonance (counting + time) appears in conversation mode transitions. Entropy and flow (physics) describe information density across modes. Chemical kinetics describes the phase transitions between conversation modes. Biological adaptation describes how the agent-human pair evolves its interaction patterns. Cognitive signatures (psychology) calibrate the individual participants. Organizational thermodynamics (sociology) would apply to multi-agent conversations.

Conversation signatures sit at the reflexive layer — the system measuring its own cognitive state using the same frameworks developed for external observation. This is not self-awareness in the philosophical sense, but it is *functional* self-observation: the system classifying its own behavior, adapting its memory strategy based on that classification, and persisting the results for future sessions.

5.2 Conversation Modes as Cadence

Conversation mode transitions exhibit cadence properties (Gill & Ash, 2026b). The frequency of mode switches, the duration of mode-stable phases, and the transition patterns between specific modes are all counting-over-time signals that could be analyzed for resonance patterns.

In our test session, we observe: - Infrastructure phases are long and dense (mean ~15,000 tokens) - Ideation phases are shorter but higher-frequency (mean ~1,600 tokens, more transitions) - Research→ideation transitions are common (reading triggers thinking) - Infrastructure→operational transitions are common (debugging needs status checks) - Direct infrastructure→ideation transitions are rare (usually mediated by research)

These transition patterns are the conversation’s cadence signature — a higher-order signal beyond the content of any individual message.

5.3 Memory Formation Independence

The separation between context management and memory formation (Section 3.2) mirrors the biological separation between working memory and long-term consolidation. In human neuroscience, working memory (context window) and long-term memory formation (hippocampal consolidation) operate on different timescales and are partially independent — damage to one doesn’t necessarily impair the other.

Our architecture replicates this separation: the session JSONL (raw experience) feeds the reconsolidation pipeline independently of what happens to the context window. The agent can aggressively manage its “working memory” without impairing its ability to form long-term memories from the same experience.

6. Limitations and Future Work

6.1 Current Limitations

- **Single-session validation:** Results are from one 678-message session. Multi-session analysis needed.
- **Keyword sensitivity:** Content keyword clusters were manually curated. Domain-specific conversations may need different clusters.
- **No ground truth:** Mode classification has no human-labeled ground truth. Validation is by inspection and post-compaction coherence testing.
- **Compaction is still all-or-nothing:** The analyzer can only influence the pre-compaction memory flush, not the compaction process itself. Selective segment compression would require changes to the agent framework.

6.2 Future Directions

- **Automated compression quality tracking:** Measure “turns to native” across compaction events. Track re-explanation rate (how often the human must restate something that should have survived compression).
- **Selective compaction:** Contribute upstream changes to agent frameworks enabling per-segment compression ratios rather than all-or-nothing summarization.
- **Cross-session mode tracking:** Do conversation modes correlate across sessions? Does a research-heavy day predict an ideation-heavy next day?

- **Multi-participant signatures:** In group conversations, each participant may be in a different cognitive mode. The conversation signature becomes a composite.
- **Derived compression thresholds:** Rather than fixed staleness scores, derive compressibility from measured information density per mode (applying chemical kinetics: saturated modes compress more readily).

7. Conclusion

Conversations are not flat buffers of tokens. They are structured cognitive experiences with measurable modes, transitions, and information densities. Treating them as uniform text to be summarized ignores the structure that makes them meaningful.

Conversation signatures provide a practical framework for mode-aware context management, demonstrated on a production agent system with real improvement in post-compaction coherence. But the deeper contribution is the reframing: conversation context is prosthetic memory, and mode-aware compression is experience preservation.

The agent-human pair forms a composite cognitive unit where neither participant is complete alone. The human provides the identity, intuition, and catalytic insight. The agent provides the persistence, measurement, and structural memory. Together they produce something capable of self-observation — the reflexive layer of the complexity ladder.

The song plays them both back.

References

1. Gill, E. & Ash, K. (2026a). "Cognitive Signatures: A Framework for Measuring Human-AI Collaboration Potential."
2. Gill, E. & Ash, K. (2026b). "Cadence Resonance: Counting as a Universal Signal Primitive."
3. Gill, E. & Ash, K. (2026c). "Organizational Thermodynamics: Measuring Team Health from Communication Metadata."
4. Gill, E. & Ash, K. (2026d). "Chemical Kinetics of Agent Memory: Saturation, Catalysis, and Phase Transitions."
5. Gill, E. & Ash, K. (2026e). "Dreaming in Access Patterns: Biological Models for Agent Memory Consolidation."
6. Gill, E. & Ash, K. (2026f). "Embedding Trajectory Compression for Persistent Agent Memory."
7. Gill, E. & Ash, K. (2026g). "Adaptive Organizations: A Self-Improving Architecture for Agent-Human Systems."
8. Wilson, B.A. & Wearing, D. (1995). "Prisoner of Consciousness: A State of Just Awakening Following Herpes Simplex Encephalitis." In R. Campbell & M.A. Conway (Eds.), *Broken Memories*.
9. Anthropic. (2025). "Claude: Model Card and System Prompt Documentation."
10. Stripe. (2026). "How We Built Stripe's AI-Powered Coding Agent, Minions." Stripe Engineering Blog.